3. Survey Report on Emerging Trends in Low-Code Application Development

Aim

To study and analyse the emerging trends in low-code application development and to understand their effect on modern software development.

Theory

Low-code platforms are changing quickly as organisations push for faster digital transformation. The trends found in the survey are summarised below.

Trend	What it means	Effect on development
AI-assisted building	Assistants suggest fields, scripts, charts and whole applications from a description	Faster start; output still has to be checked by the developer
Citizen development	Business users build and maintain departmental applications	Less backlog for IT; needs governance
Cloud-native delivery	Platforms are hosted in the cloud with built-in scaling and updates	No servers to manage; easy collaboration
Process automation	Workflows, approvals and schedules replace repetitive manual steps	Fewer errors and delays
Multi-experience apps	One application runs on web and mobile	Lower cost of mobile support
API and integration	Connectors and REST calls link CRM, databases and cloud services	Low-code fits into enterprise landscapes
Security and governance	Role-based access, audit logs, encryption and compliance tooling	Safer enterprise use
DevOps for low-code	Environments, versioning and release pipelines	Controlled deployment and rollback

Procedure

1. Study the concept of low-code application development.
2. Identify popular low-code platforms and their key features.
3. Collect information on current trends from research papers, analyst reports and vendor documentation.
4. Analyse each trend: AI assistance, citizen development, cloud delivery, automation, integration, security and DevOps.
5. Compare the benefits and the challenges of each trend.
6. Record the observations and prepare the survey report.

Result

The survey shows that low-code development is moving towards AI-assisted building, cloud-native platforms, citizen development, process automation, easy integration and stronger governance. These trends shorten delivery time and lower cost, while security, governance and customisation limits remain the main challenges.